Youngwoong Cho

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Education

- **Georgia Institute of Technology** *Expected May 2028*
  M.S. in Robotics Atlanta, GA
- **The Cooper Union for the Advancement of Science and Art** *Class of 2023*
  B.E. in Mechanical Engineering; Minor in Computer Science New York, NY
  - **Summa Cum Laude** | GPA: **3.92/4.00** | Engineering GPA: **3.96/4.00**
  - Norman Perry Internship Fund | Half Tuition Scholarship | Innovator's Merit Award | William C. and Esther Hoffman Beller Prize

Research Experience

- **Robot Learning and Reasoning Lab (RL2), Georgia Tech** *August 2026 – Present*
  Graduate Researcher | Prof. Danfei Xu Atlanta, GA
  - **Learning Mobile Manipulation from Human Demonstrations:** Training bimanual policies for an RBY1 humanoid using only egocentric human demonstrations, and studying which observation choices (viewpoint alignment, point clouds vs. RGB, removing the human hand) close the human-to-robot gap
  - **Sim-to-Real Policy Screening:** Built a simulation replica of the real-robot task to compare human-trained checkpoints and diagnose real-robot failures before spending robot time
  - **Cross-Embodiment Dexterous Manipulation:** Built an Apple Vision Pro–Isaac Sim teleoperation pipeline and a shared wrist–fingertip action space for 7 dexterous hands, and studied whether co-training on more hand types improves zero-shot transfer to unseen hands
  - **Training and Evaluation Platform:** Built a platform that compiles reproducible Slurm jobs from versioned adapters across two GPU clusters, with checkpoint auto-resume, multi-GPU DDP trainers, simulation evaluation suites (LIBERO, DexJoCo, MuJoCo, Isaac Sim), real-robot RB-Y1 rollout evaluation, and live Vision Pro teleoperation for data collection
- **RLWRLD (Robotics Foundation-Model Startup)** *April 2026 – August 2026*
  Research Engineer Seoul, Republic of Korea
  - **Fine-tuned GR00T N1.6 on DexJoCo**, an 11-task dexterous-hand benchmark, reaching **46.2%** mean success vs. 40.3% reported for N1.5; also evaluated released $\pi_{0.5}$ checkpoints
  - **Improved robot-state tokenization in the GR00T N1.6 VLA**, raising humanoid simulation success from 36.4% to **41.5%** by encoding state as 11 tokens instead of 1
  - **Adapted a 3D geometry-aware policy to a 16-DoF dexterous hand**, fixing a step-counting bug in its trainer and raising success on a single-arm task from 11% to ∼**55%**
  - **Built a training/evaluation platform on 3 compute clusters** and made MuJoCo evaluation bit-reproducible, eliminating run-to-run outcome changes in **22.5%** of same-seed episodes
- **Naver Labs** *March 2022 – August 2022*
  Deep Learning Research Intern, Autonomous Driving | Siyeong Lee Seongnam, Republic of Korea
  - **Improved 3D bounding-box prediction** with a self-supervised iterative refinement module integrated into MuZero, increasing task success from 10% to **30%**
  - **Identified RL exploration inefficiency** and redirected the learning framework toward GAIL-based imitation learning inspired by IQ-Learn
  - **Built a 3D annotation tool** for collecting human demonstrations, reducing annotation time by **70%**
- **Seoul Robotics** *May 2021 – March 2022*
  Machine Learning Research Engineer Intern | Dr. Hong Minh Truong Seoul, Republic of Korea
  - **Developed vectorized curb detection** for Seoul-specific driving environments, establishing the experimental baseline for subsequent drivable-area research
  - **Built a multi-view LiDAR annotation system** with temporal stacking and synchronized editing, producing labels that improved drivable-area detection by **15%**
  - **Standardized perception evaluation** across mAP, MOTA, and velocity metrics over **6,000+ frames**, producing the company's primary evaluation tool

- **Seoul National University** *May 2020 – July 2020*
  Research Intern, Interactive & Network Robotics Lab | Prof. Dongjun Lee Seoul, Republic of Korea
  - **Validated cooperative manipulation algorithms** on a physical multi-robot testbed, identifying and debugging systematic sim-to-real gaps

Industry Experience

- **Enmight (Subsidiary of ESU Farm LLC)** *March 2024 – December 2025*
  ERP Software Engineer Englewood Cliffs, NJ
  - **Built a closed-loop IoT monitoring system** integrating temperature, humidity, and 3-axis accelerometer sensors for real-time supply-chain quality control
  - **Implemented accelerometer sensor fusion** for state estimation, frequency-domain transport-anomaly detection, and impact-event classification, alongside thermal PID control
- **LG CNS** *July 2023 – February 2024*
  Data Analyst, Customer Data Platform Englewood Cliffs, NJ
  - **Designed a unified analytics warehouse** in GCP BigQuery integrating Google Analytics, CRM, and internal data, reducing data-model redundancy by **80%**

Selected Projects

- **Self-Driving Delivery Robot** *Spring 2023*
  Senior Capstone, The Cooper Union
  - **Built an end-to-end autonomous navigation stack** (3D LiDAR SLAM, localization, planning, control) for an indoor delivery robot, with localization latency under **100 ms**
- **BarlowZero: Sample-Efficient Model-Based Reinforcement Learning** *Fall 2022*
  ECE-472: Deep Learning, The Cooper Union
  - **Developed a MuZero variant with Barlow Twins self-supervision**, cutting training batch size from 512 to **64** while performing comparably to EfficientZero on Atari Breakout
- **Cooper Union IGVC** *Fall 2022 – Spring 2023*
  Intelligent Ground Vehicle Competition
  - **Developed LiDAR- and RGB-based localization, mapping, and path planning** in ROS, plus an OCR-based stop-sign classifier
  - **Placed 2nd in autonomous navigation** and **3rd in vehicle design**

Technical Skills

- **Robot Learning:** VLAs (GR00T N1.5/N1.6, $\pi_{0.5}$), Diffusion Policy, HAT, HPT, Imitation Learning, RL
- **Robotics:** Dexterous Manipulation, Retargeting (FK/IK), Teleoperation, Mobile Manipulation, SLAM
- **ML / Simulation:** PyTorch, FSDP, CUDA, Isaac Sim, MuJoCo, Weights & Biases
- **Programming:** Python, C++, TypeScript, JavaScript
- **Infrastructure:** Slurm, Kubernetes, ROS, Git, GCP BigQuery

Activities & Memberships

- **Tau Beta Pi**, Member *November 2020 – Present*
- **Korean Association of Cooper Union**, Vice President *August 2020 – May 2021*